M. Sc. Melanie Maier

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Date of Birth: 2002-03-02

EDUCATION

Technical University of Munich (TUM), Munich - *M. Sc. Mathematics in Data Science*

April 2023 - October 2025

Focus: Machine Learning and Statistics

Thesis: "A Systematic Evaluation of RRI and Morphological Wavelet Features for PVC Detection in ECG Signals Using Machine Learning" (Supervisor: Prof. Dr. Daniel Rückert)

Université de Strasbourg (Unistra), Strasbourg - *Erasmus+ Semester*

August 2024 - January 2025

Coursework: Statistical Learning, Advanced Data Analysis, Data Mining, Generalized Linear Models, C++

Technical University of Munich (TUM), Munich - *B. Sc. Mathematics*

October 2019 - March 2023

Focus: Analysis, Functional Analysis, Partial Differential Equations

Thesis: "The Hille-Yosida Theorem for Semigroups of Linear Operators" (Supervisor: Prof. Dr. Daniel Matthes)

RESEARCH EXPERIENCE

Munich Center for Machine Learning - *Incoming PhD Student*

August 2026 - August 2029

TUM Chair of Computational Imaging and AI in Medicine - *Incoming PhD Student*

August 2026 - August 2029

- Prospective PhD topic: 3D reconstruction from medical ultrasound imaging data

TUM Klinikum rechts der Isar, Munich - *Research Assistant*

November 2025 - Present

- Optimized classical ML and deep learning algorithms for PVC detection in long-term ECG recordings to support clinical translation of my Master's thesis work (scientific manuscript under review for MICCAI)
- Explored scalable long-sequence Transformer designs for 24-hour ECG data, addressing context length, memory constraints, and long-range dependency modeling

TU Darmstadt - *Collaborative Research Project*

August 2025 - January 2026

- Generated counterfactual and unseen test scenarios in the JAXAtari framework to train and evaluate a vision-language model assisted reinforcement learning agent, improving robustness and generalization in novel environments (scientific manuscript under review for ICML)

TUM Data Innovation Lab, Munich - *Research Project Data Science in Earth Observation*

October 2024 - March 2025

- Adapted an Elucidated Diffusion Model (EDM) implementation to be trained on satellite imagery for sea surface temperature image downscaling

TUM Klinikum rechts der Isar, Munich - *Student Research Assistant Biosignalanalysis*

May 2022 - July 2024

- Supported the discovery of novel medical correlations in large scale medical studies (KORA, NAKO) for example among diabetes and respiratory rate
- Presented a novel wavelet-transform based ML-algorithm to detect cardiac arrhythmia (PVC) in an ECG signal, scoring an accuracy of > 99.2% on a balanced dataset with more than 1,000,000 heartbeat samples (NAKO)

SCIENTIFIC PUBLICATIONS

- I.-M. Rückert-Eheberg, A. Steger, A. Müller, ..., M. Maier, et al. (2025): Respiratory rate and its associations with disease and lifestyle factors in the general population – results from the KORA-FF4 study. PLOS ONE.
- A. Steger, P. Barthel, A. Müller, ..., M. Maier et al. (2024): Deceleration capacity derived from a five-minute electrocardiogram predicts mortality in the general population. Scientific Reports (Nature Portfolio).
- E. Martens, H.-U. Haase, G. Mastella, ..., M. Maier et al. (2024): Smart hospital: achieving interoperability and raw data collection from medical devices in clinical routine. Frontiers in Digital Health.

INDUSTRY EXPERIENCE

Intrinsic, Munich (now Google) - *Data Science for Robot Operations Intern*

April 2025 - September 2025

- Improved post-commissioning workflows and accelerated performance monitoring by implementing a scalable metrics service using gRPC to handle the collection and computation of key performance indicators such as Overall Equipment Effectiveness

CHECK24 (Kredite24), Munich - *Data Science Working Student*

October 2024 - March 2025

- Improved the detection of important metrics by up to 5% F1 score in a low quality document image by implementing a line detection algorithm based on the bounding box rotation of words
- Supported NLP model development and evaluation, including named entity recognition with LSTM and xLSTM (Tensorflow) and experimented with computationally efficient large language models.

Rocket Factory Augsburg, Augsburg - *Software Engineering, Mission Analysis Intern*

August 2024 - September 2024

- Enabled the mission analysis team to make flight-critical decisions by implementing a Python software to visualize a given rocket trajectory and its visibility to a selection of ground stations
- Improved rocket trajectory optimization by implementing an initial guess creator using a supervised learning approach

EY, Munich - *IT-Consulting Cybersecurity Intern*

August 2021 - September 2021

- Strengthened clients' cyber resilience by conducting ITGC testing on backup and recovery processes, ensuring compliance with audit and regulatory standards

PERSONAL SOFTWARE AND DATA PROJECTS

Equine Care Collaboration Platform (MVP), Munich - Private Project

October 2025 -December 2026

- Developing an early-stage cloud-based collaboration and documentation platform for equine care teams (owners, veterinarians, therapists), enabling shared case history, media upload, and structured treatment notes with a focus on workflow design, role-based access, and secure document sharing

AI-Based Knowledge Management (MVP), Munich - Participant in TUM.ai *E-Lab 4.0*

October 2025 -December 2026

- Building an MS Teams bot that attends meetings and auto-creates documentation from transcripts and screen recordings where knowledge is managed within a RAG pipeline

Computer Vision-Based Sports Coaching Tool, Munich - Private Project

February 2025 -September 2025

- Enhancing athlete learning and coaching efficacy in equestrian sports by developing a posture analysis tool based on keypoint detection, combined with an algorithmical movement interpretation and personalized language feedback in a team of two people: www.posequest.com

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AWARDS AND HONORS

- **TUM.ai Entrepreneurship Lab** - Member, October 2025 - January 2026
- **MDSI/Linde Master Scholarship** - Scholarship Recipient, 2023 - 2024
- **Deutsche Physikalische Gesellschaft (DPG)** - Abitur Graduation Prize, June 2019
- **Landeswettbewerb Mathematik** - 2nd Prize, 2016
- **Landeswettbewerb Schüler experimentieren** - 3rd Prize, 2014